

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-13: Color Image Processing

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-6)

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2.5.3 The HSI Color Model

As we have seen, creating colors in the RGB and CMY models and changing from one model to the other is a **straightforward process**.

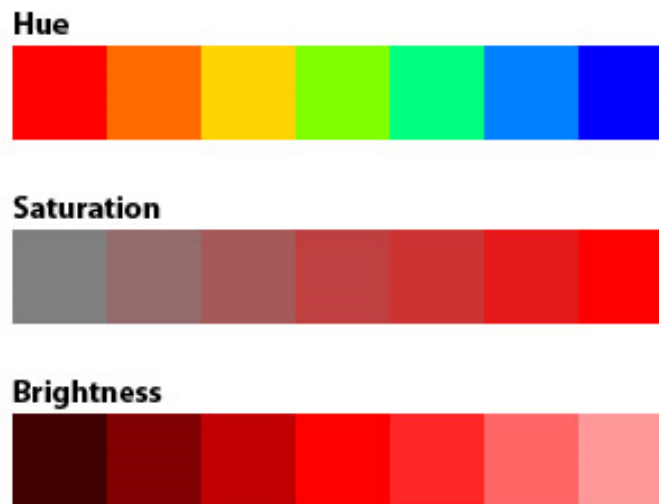
As noted earlier, these color systems are ideally suited for hardware implementations.

*** Unfortunately, the RGB, CMY, and other similar color models are not well suited for describing colors in terms that are practical for human interpretation.

When humans view a color object, we describe it by its hue, saturation, and brightness.

Hue is a color attribute that describes a pure color (pure yellow, orange, or red), whereas **saturation** gives a measure of the degree to which a pure color is diluted by white light.

Brightness is a subjective descriptor that is practically impossible to measure. It embodies the achromatic notion of intensity and is one of the key factors in describing color sensation.



Why HSI model?

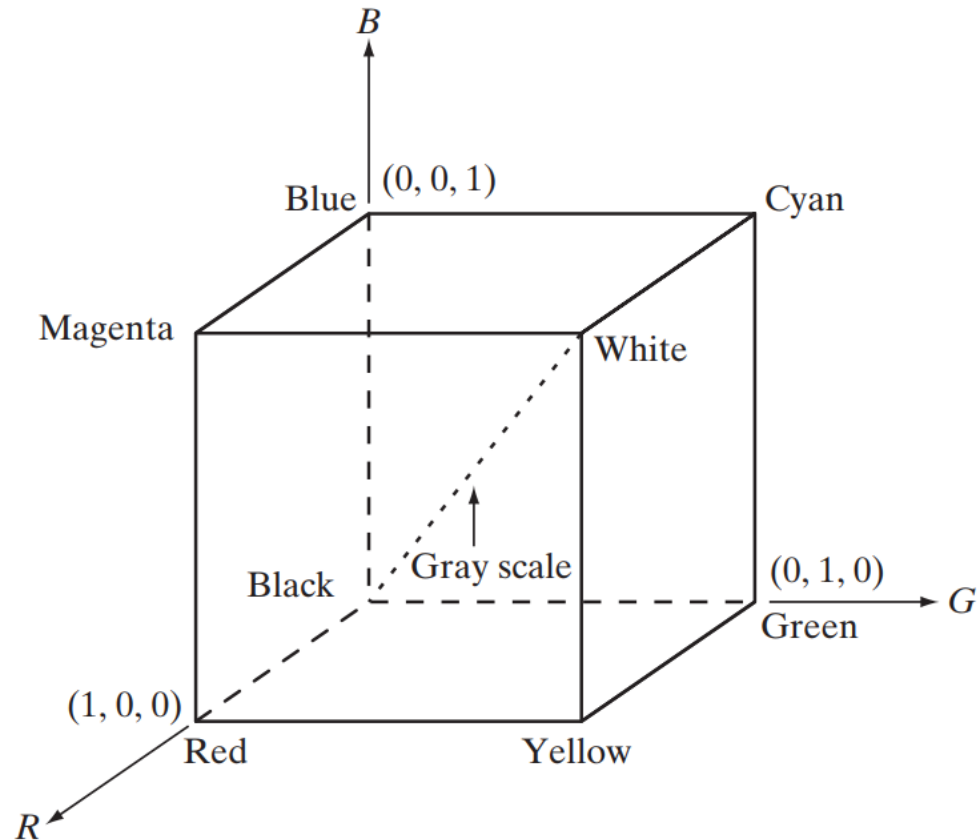
We do know that intensity (gray level) is a most useful descriptor of monochromatic images. This quantity definitely is measurable and easily interpretable.

*** The model we are about to present, called the HSI (hue, saturation, intensity) color model, decouples the intensity component from the color-carrying information (hue and saturation) in a color image.

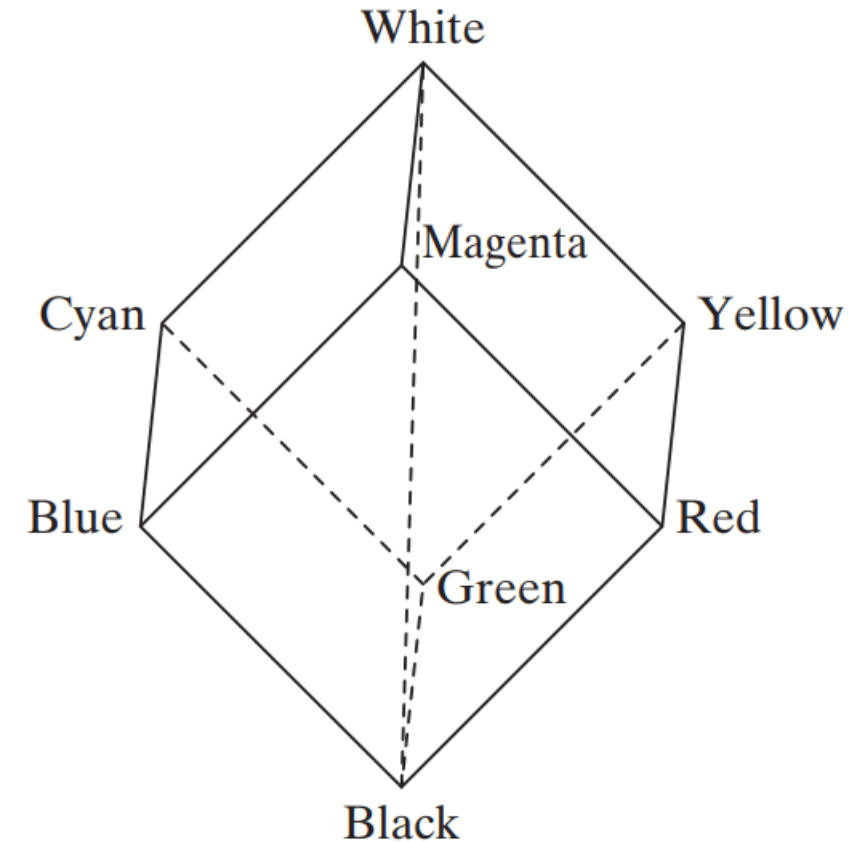
- ✓ As a result, the HSI model is an ideal tool for developing image processing algorithms based on color descriptions that are natural and intuitive to humans, who, after all, are the developers and users of these algorithms.

Relationship between RGB and HSI

This becomes rather clear if we take the color cube and stand it on the black $(0, 0, 0)$ vertex, with the white vertex $(1, 1, 1)$ directly above it.

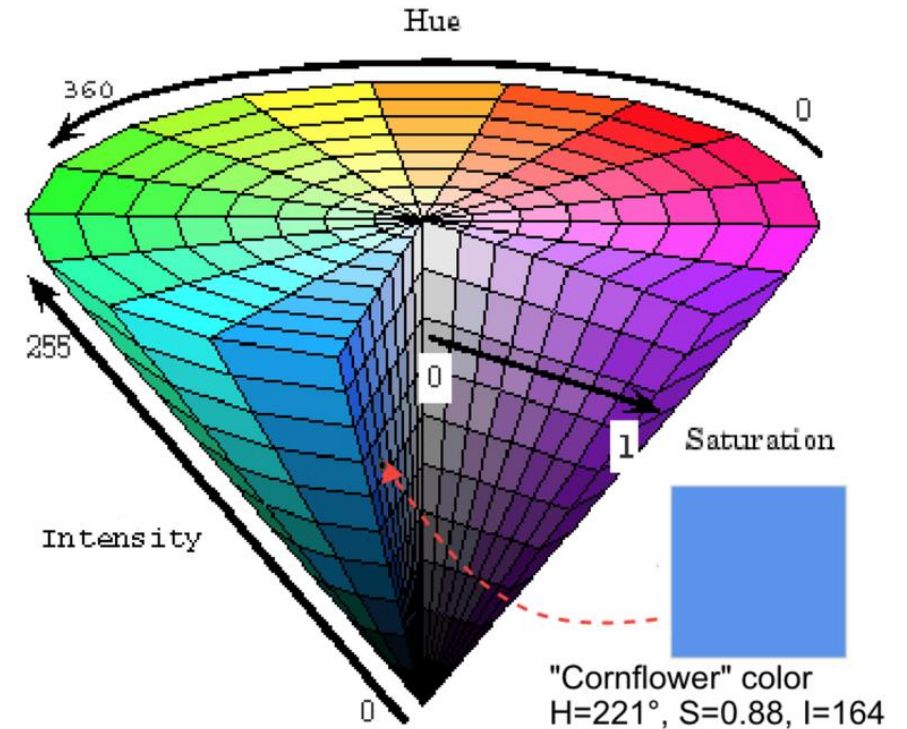
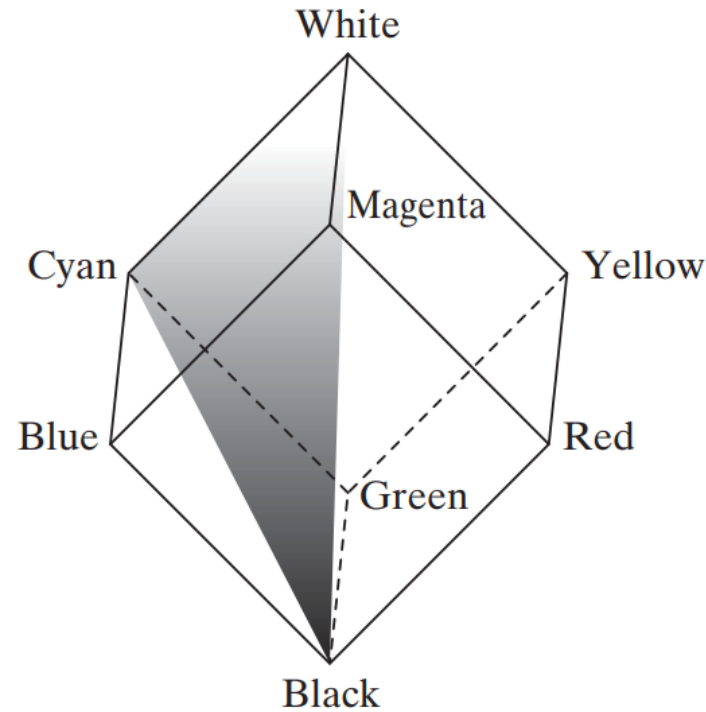
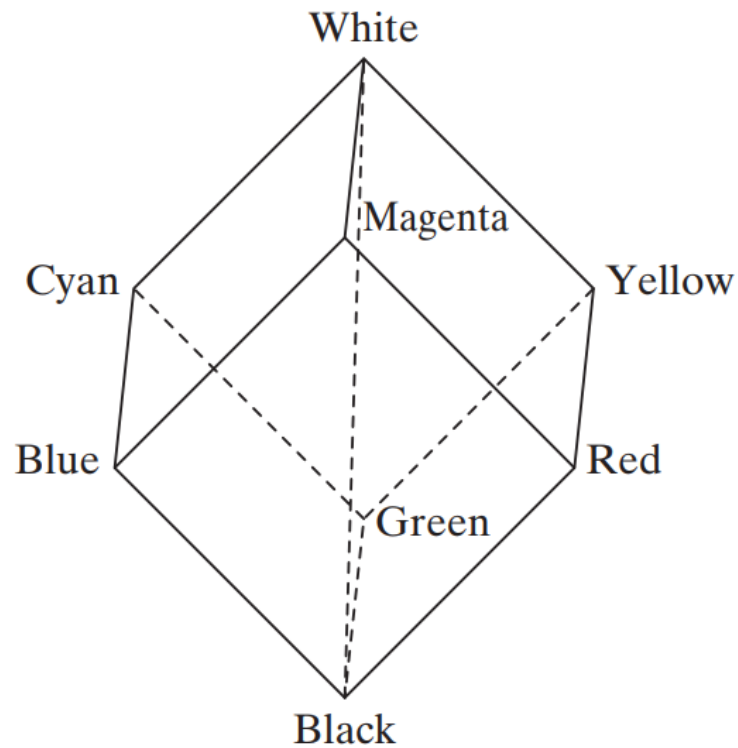


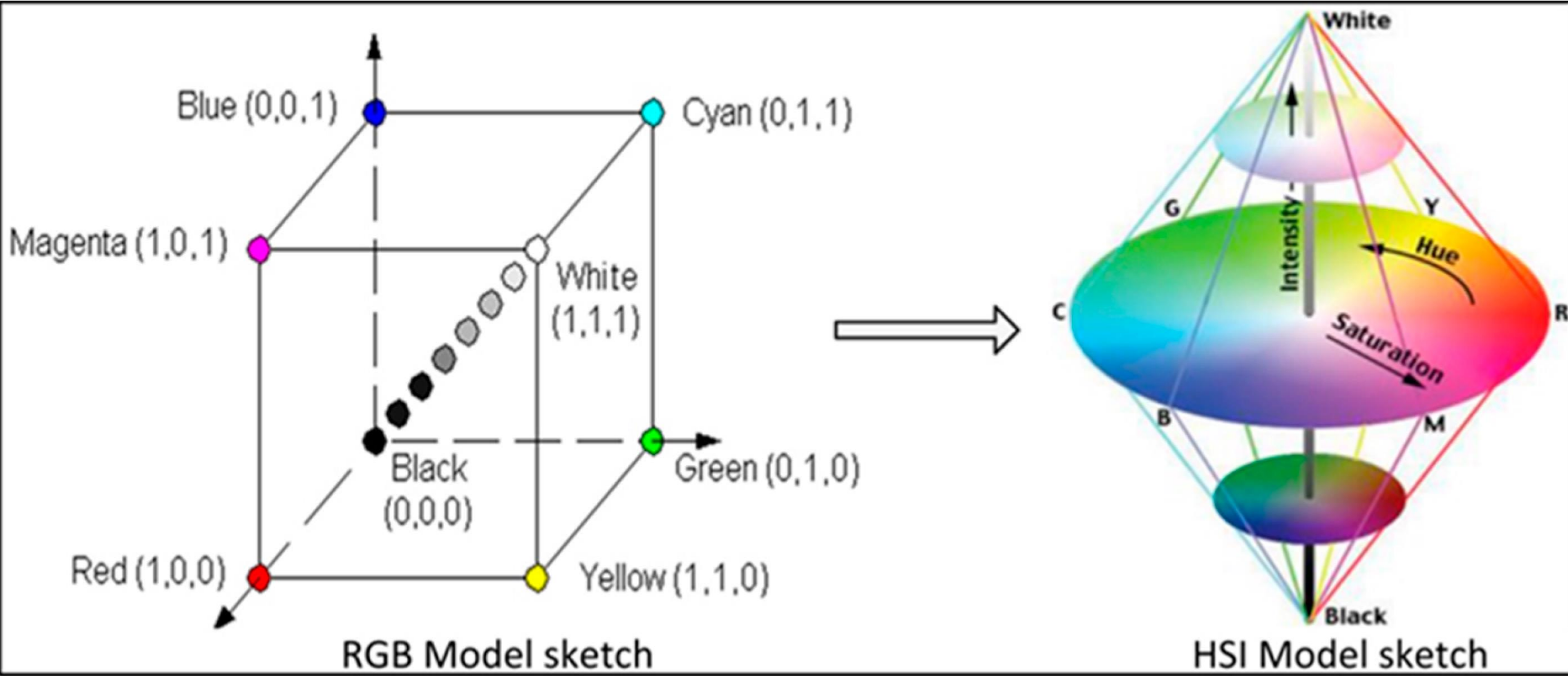
The intensity (gray scale) is along the line joining these two vertices



The line (intensity axis) joining the black and white vertices is vertical

Thus, if we wanted to determine the intensity component of any color point in Fig, we would simply pass a plane perpendicular to the intensity axis and containing the color point. The intersection of the plane with the intensity axis would give us a point with intensity value in the range $[0, 1]$.





In this plane we see that the primary colors are separated by 120° . The secondary colors are 60° from the primaries, which means that the angle between secondaries also is 120° .

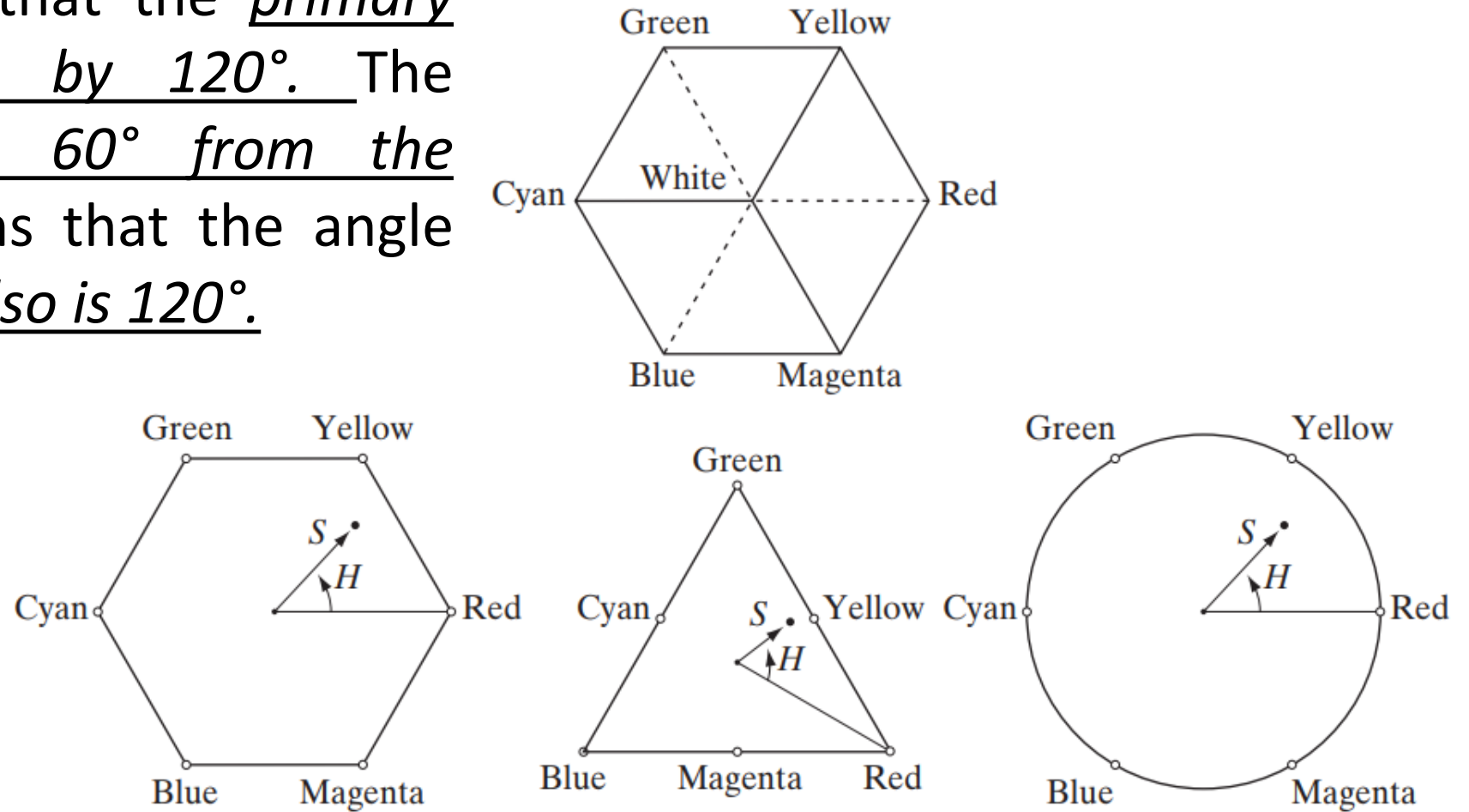


Figure: Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation.

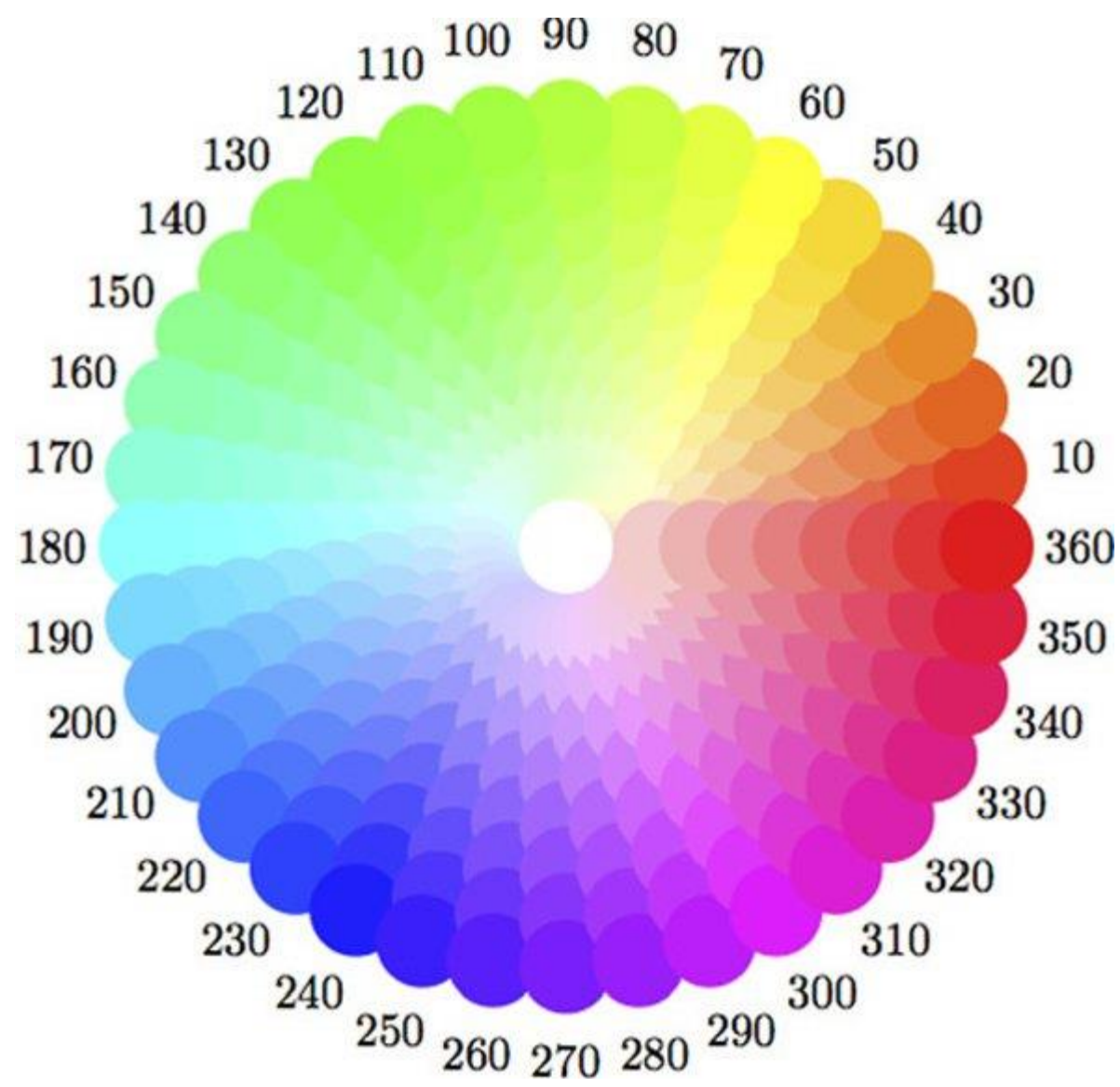
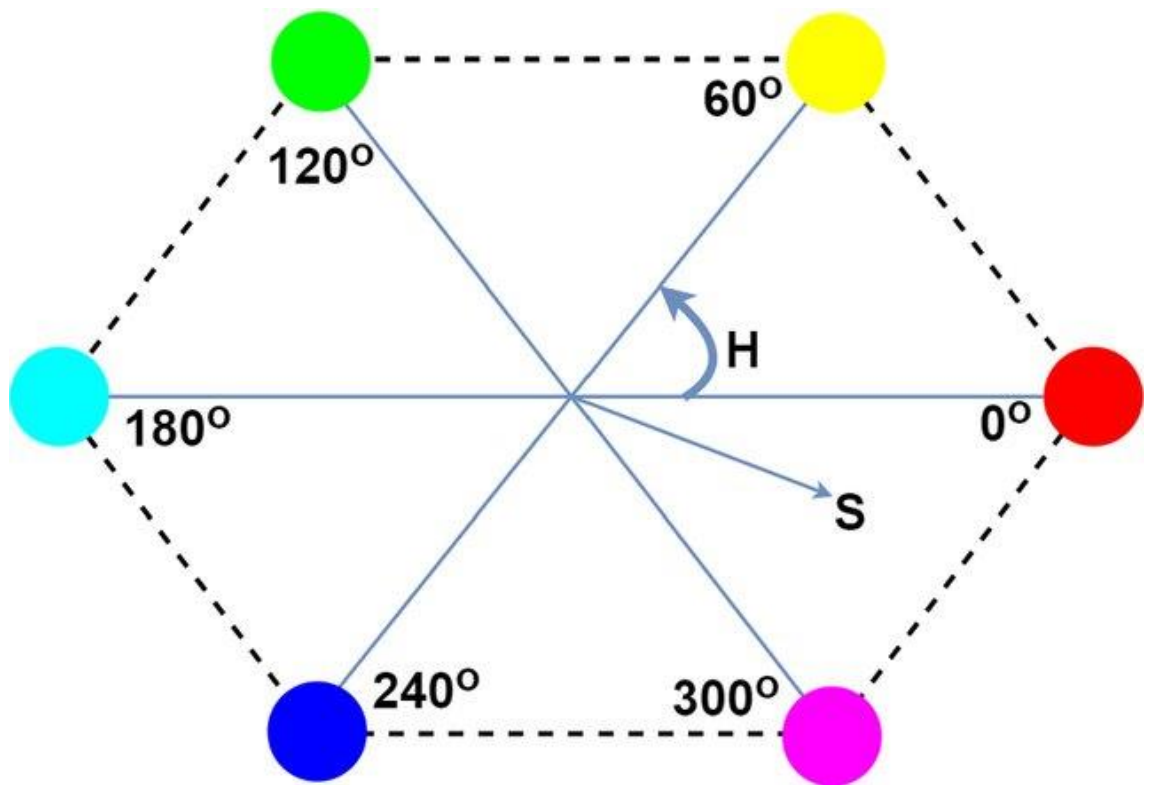
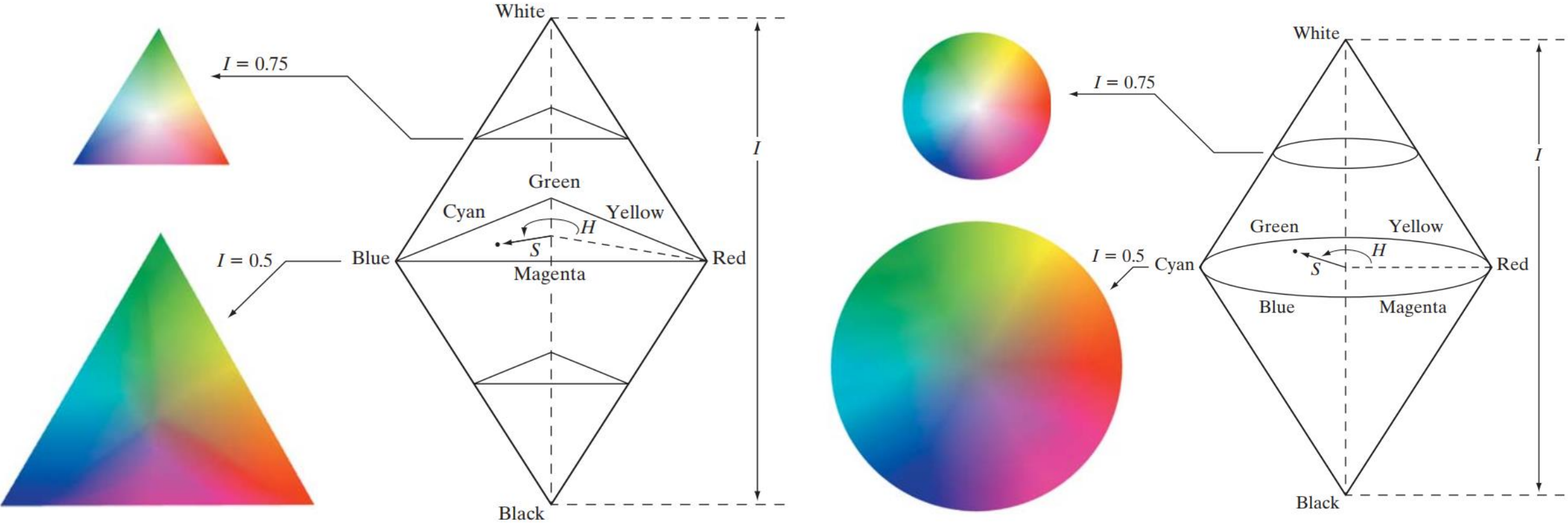


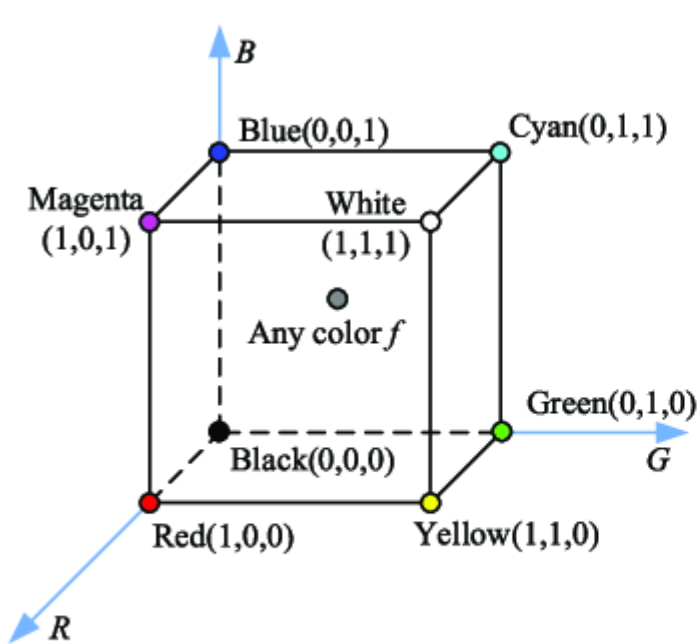
Figure shows the HSI model based on color triangles and also on circles.



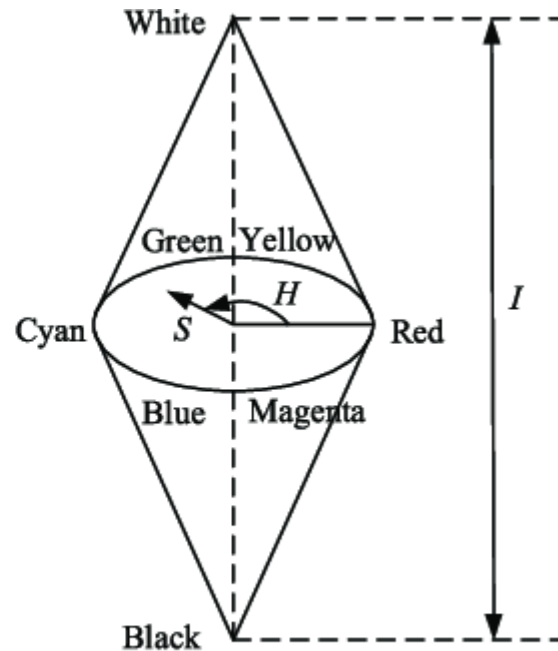
2.5.3.1 Converting colors from RGB to HSI

Given an image in RGB color format, the H component of each RGB pixel is obtained using the equation

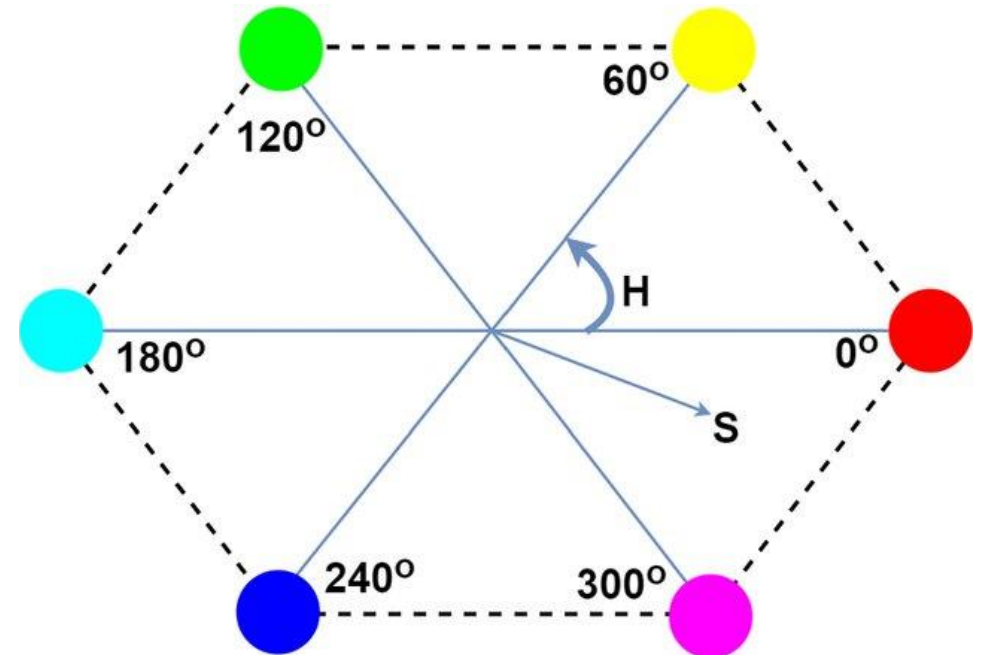
$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases} \quad \theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$



(a) Normalized RGB color space model



(b) HSI space model



The saturation component is given by $S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$

Finally, the intensity component is given by $I = \frac{1}{3}(R + G + B)$

It is assumed that the RGB values have been normalized to the range [0, 1] and that angle is measured with respect to the red axis of the HSI space.

- Hue can be normalized to the range [0, 1] by dividing by 360° all values resulting from Eq.

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

- The other two HSI components already are in this range if the given RGB values are in the interval [0, 1].

Steps of Conversion

Normalize the RGB Values:

First, convert the RGB values to the $[0, 1]$ range by dividing each component by 255. For example, consider a pixel with RGB values (100, 150, 200):

$$R = \frac{100}{255} \approx 0.392$$

$$G = \frac{150}{255} \approx 0.588$$

$$B = \frac{200}{255} \approx 0.784$$

Calculate Intensity (I):

The Intensity component is the average of the normalized RGB values:

$$I = \frac{R+G+B}{3} = \frac{0.392+0.588+0.784}{3} \approx 0.588$$

Calculate Saturation (S):

Saturation measures the purity of the color. It's calculated as:

$$S = 1 - \frac{3 \times \min(R, G, B)}{R+G+B}$$

$$\min(R, G, B) = \min(0.392, 0.588, 0.784) = 0.392$$

$$S = 1 - \frac{3 \times 0.392}{0.392+0.588+0.784} \approx 0.333$$

Calculate Hue (H):

Hue represents the color type and is calculated using:

$$\theta = \arccos \left(\frac{0.5 \times [(R-G) + (R-B)]}{\sqrt{(R-G)^2 + (R-B)(G-B)}} \right)$$

$$\theta = \arccos \left(\frac{0.5 \times [(0.392 - 0.588) + (0.392 - 0.784)]}{\sqrt{(0.392 - 0.588)^2 + (0.392 - 0.784)(0.588 - 0.784)}} \right)$$

$$\text{numerator} = 0.5 \times (-0.196 - 0.392) = -0.294$$

$$\text{denominator} = \sqrt{(-0.196)^2 + (-0.392)(-0.196)} \approx 0.196$$

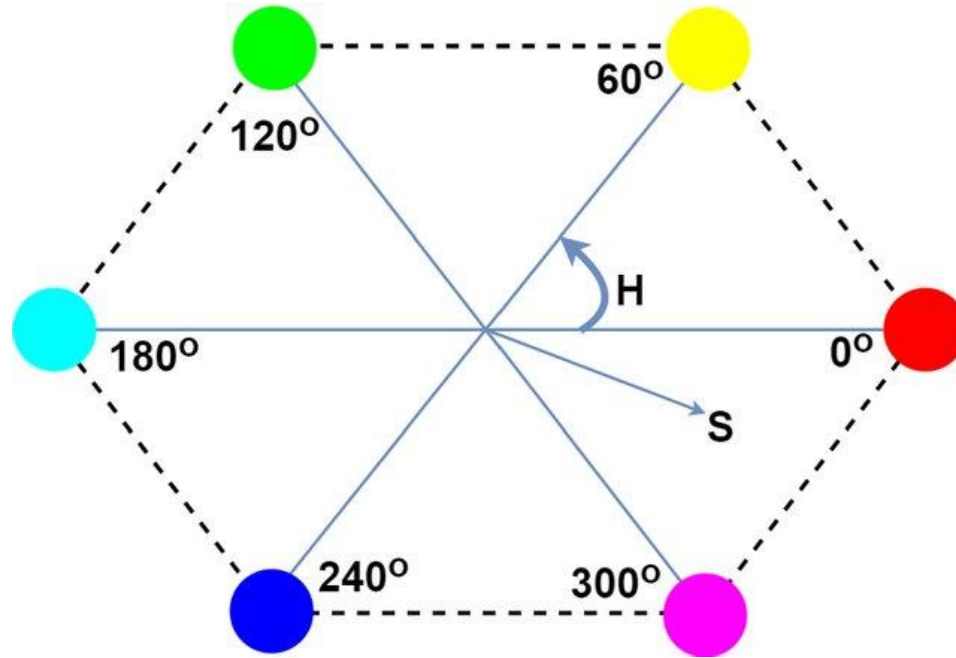
$$\theta = \arccos \left(\frac{-0.294}{0.196} \right)$$

if $B \leq G$, then $H = \theta$; otherwise, $H = 360^\circ - \theta$.

Finally, normalize H to the $[0, 1]$ range by dividing by 360° .

The cosine inverse function $\cos^{-1}$ always returns values between 0° and 180°

However, the full hue range is 0° to 360° , so we need an adjustment for cases where the hue falls in the second half of the color wheel (180° to 360°).



If B is greater than G , it means the hue angle is in the lower hemisphere of the color wheel, so we subtract it from 360° to place it correctly.

2.5.3.2 Converting colors from HSI to RGB

Understanding the HSI Color Model:

Hue (H):

Represents the type of color and is typically measured in degrees (0° to 360°), corresponding to colors like red, green, blue, etc.

Saturation (S):

Indicates the vibrancy of the color, ranging from 0 to 1. A saturation of 0 means the color is a shade of gray, while 1 represents a fully saturated color.

Intensity (I):

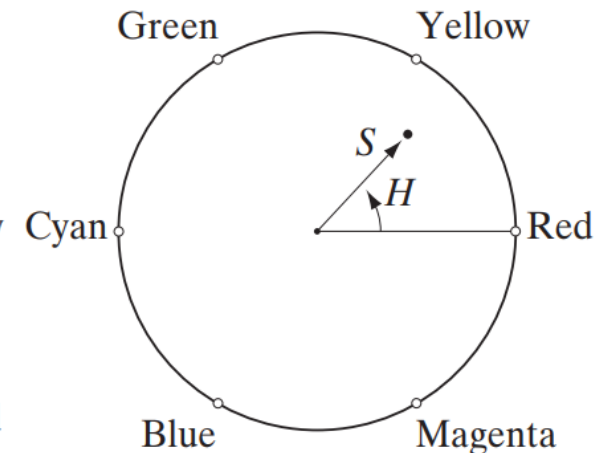
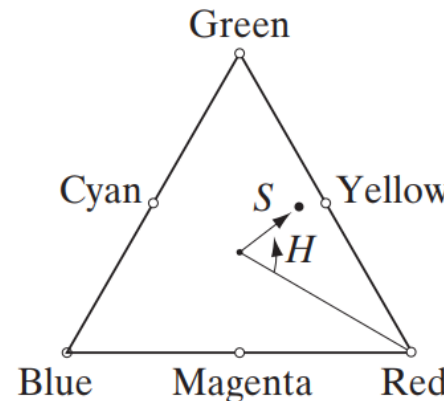
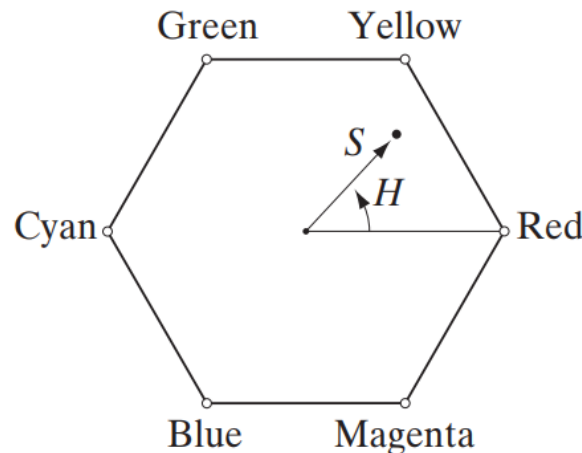
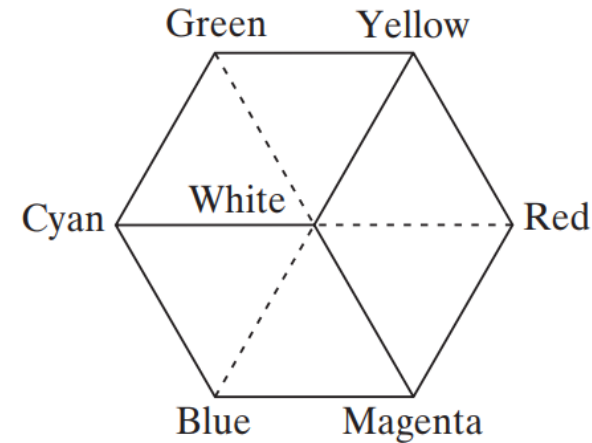
Denotes the brightness of the color, also ranging from 0 to 1.

2.5.3.2 Converting colors from HSI to RGB

Given values of HSI in the interval $[0, 1]$, we now want to find the corresponding RGB values in the same range.

The applicable equations depend on the values of H.

There are three sectors of interest, corresponding to the 120° intervals in the separation of primaries.



We begin by multiplying H by 360° , which returns the hue to its original range of $[0^\circ, 360^\circ]$.

RG sector ($0^\circ \leq H < 120^\circ$): When H is in this sector, the RGB components are given by the equations

$$B = I(1 - S)$$

$$R = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

$$G = 3I - (R + B)$$

GB sector ($120^\circ \leq H < 240^\circ$): If the given value of H is in this sector, we first subtract 120° from it:

$$H = H - 120^\circ$$

Then the RGB components are

$$R = I(1 - S)$$

$$G = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

$$B = 3I - (R + G)$$

BR sector ($240^\circ \leq H \leq 360^\circ$): Finally, if H is in this range, we subtract 240° from it:

$$H = H - 240^\circ$$

Then the RGB components are

$$G = I(1 - S)$$

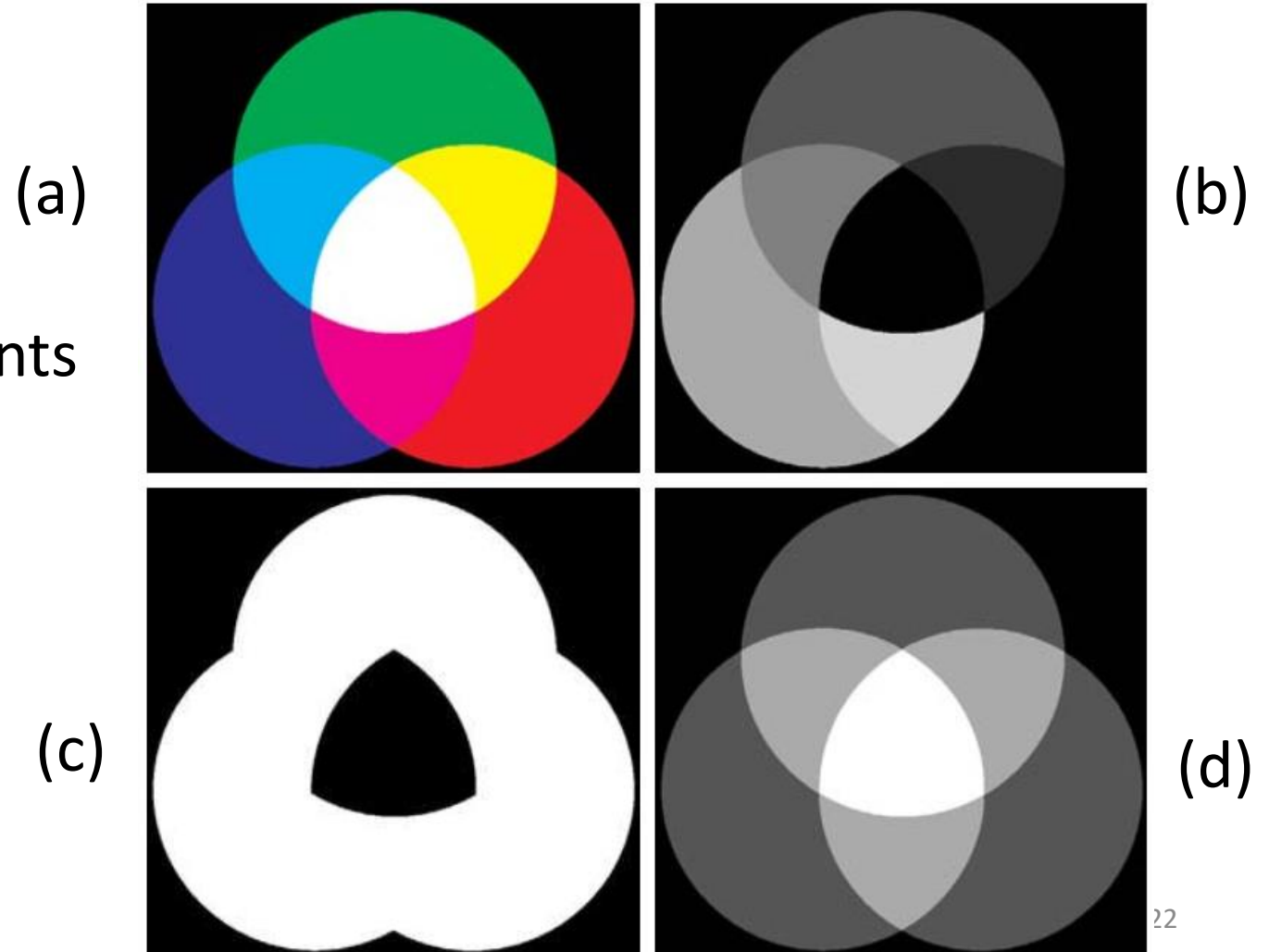
$$B = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

$$R = 3I - (G + B)$$

2.5.3.3 Manipulating HSI component images

Figure (a) shows an image composed of the primary and secondary RGB colors. Figures (b) through (d) show the H , S , and I components of this image.

(a) RGB image and the components
of its corresponding HSI image
(b) hue,
(c) saturation, and
(d) intensity.



$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

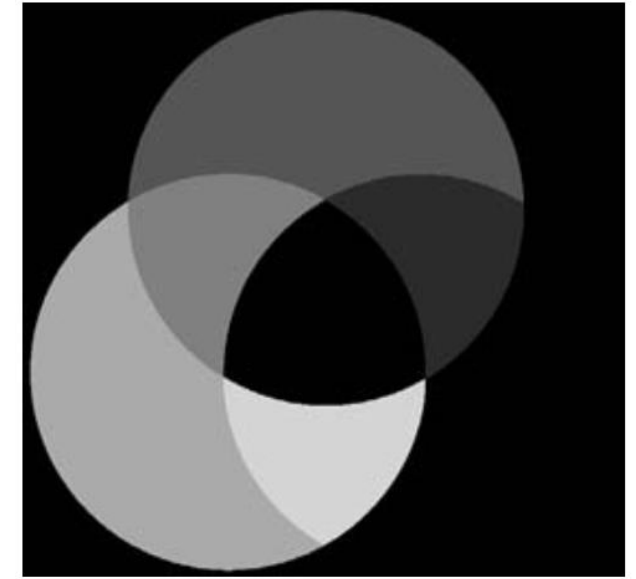
$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

$$I = \frac{1}{3}(R + G + B)$$

To change the individual color of any region in the RGB image, we change the values of the corresponding region in the **hue** image.

*** Then we convert the new H image, along with the unchanged S and I images, back to RGB.



- ❑ To change the color of a region, the hue (H) component is adjusted, and the modified HSI image is converted back to RGB.
- ❑ To change the saturation (color purity), adjustments are made to the S component, while changes in intensity affect brightness.

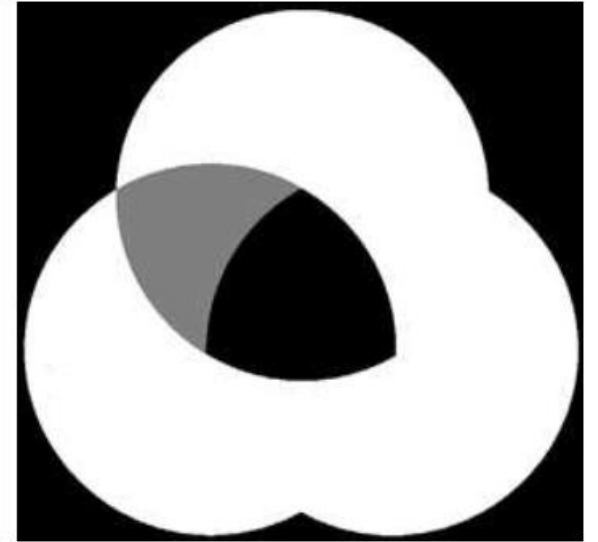
(a)–(c) Modified HSI component images.

(d) Resulting RGB image.

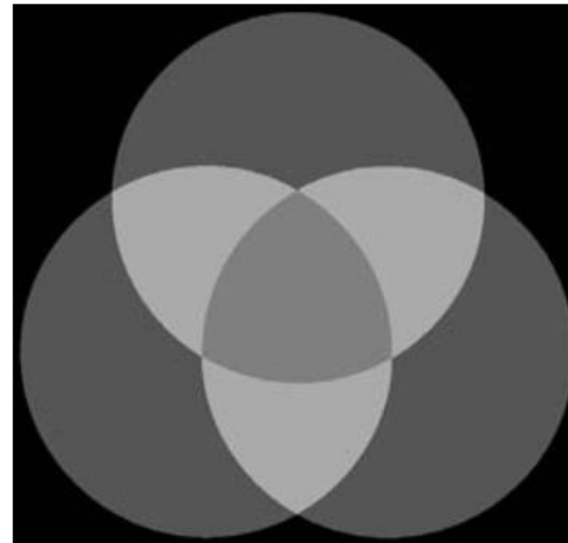
(a)



(b)



(c)



(d)

